

CSU44051 - Human Factors

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Module Aims

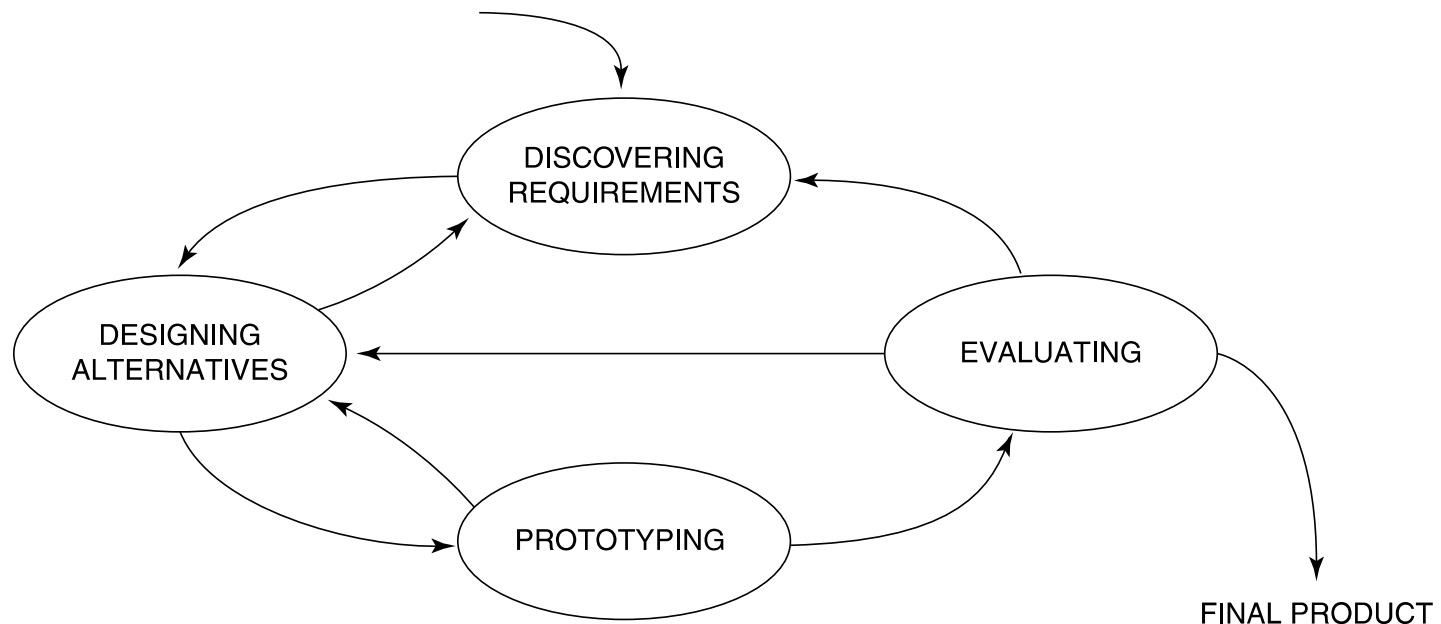
This course is an introduction to human factors and human-computer interaction (HCI).

It aims to give you:

1. A set of concepts, models for thinking about interaction between people and computers.
2. Understanding, and experience, of the human-centred design process.
3. Knowledge of more specialised subtopics such as CSCW, human performance, user engagement.

Course Structure

- This course is structured around the *design process*.
- A large part of the course (and assessment) is a group design project.



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Provisional Plan

Before Reading Week	After Reading Week
• Understanding people and interaction • Discovering Requirements • HCI Research	• Designing Alternatives • Prototyping • Evaluating • Specialised Topics

Coursework

- Continuous assessment (tutorials and in-class quizzes) 20%
- Reading assignment 10%
- Design assignment 30%
- Prototyping assignment 30%
- Group prototype presentation 10%

Lectures & Tutorials

Tutorial	Lecture	Lecture
Tuesday 9am	Wednesday 11am	Thursday 10am
E3 01.037 (large room), 01.041, 01.057 (small rooms)	LB01	LB04

- Lecture and tutorial attendance is mandatory.
- The first tutorial is Tuesday 22nd September

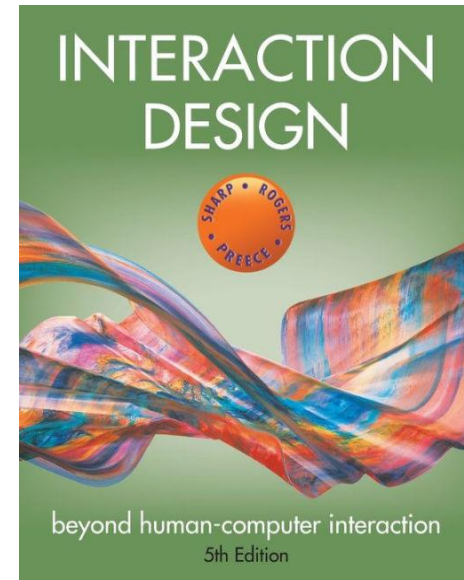
Contact

- Email Gavin.Doherty@tcd.ie
- Put CSU44051 in the subject

Course textbook

Interaction Design: Beyond Human-computer Interaction

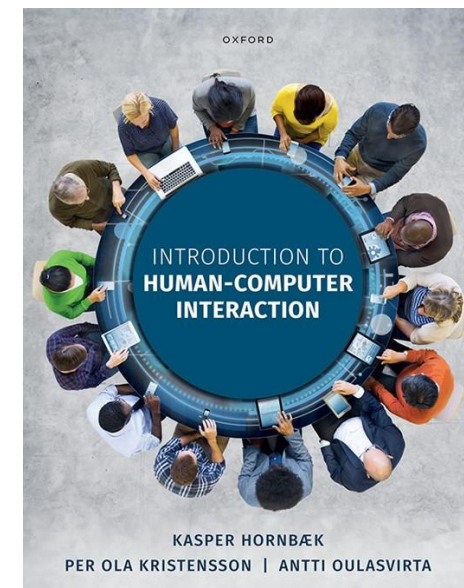
Y. Rogers, H. Sharp, J. Preece
4th (2015) or 5th (2019) or 6th (2023) Edition



Introduction to Human-Computer Interaction

E-book free from OUP

<https://academic.oup.com/book/60808>



Further Reading

- **The Design of Everyday Things**, D. Norman, MIT Press, 2013

An influential book and author. A good introduction to thinking about interaction design

- **About Face**, A. Cooper, R. Reimann, D. Cronin, C. Noessel, John Wiley and Sons, 2014

Well known textbook that details in depth an approach for personas and scenarios.

- **Human-Computer Interaction**, 3e, A. Dix, J. Finlay, G. Abowd, R. Beale, Prentice Hall, 2004, hcibook.com/e3

Old but gives a stronger engineering perspective.

Further Reading

- **Designing User Experience: A guide to HCI, UX and interaction design**, D. Benyon, 2019, Pearson. Practical focus, information systems.
- **Engineering psychology and human performance**, Wickens et al, 5th ed., 2021. Human Factors, Ergonomics, safety, aviation etc.
- **Sketching User Experiences**, B. Buxton, Morgan Kaufmann 2007. An interesting perspective/retrospective on design.

Interactive Systems

- Most computer systems interact with a user or users in some way.
- This can be directly, for example sitting in front of a PC, giving input by typing, mouse operations, and viewing output from the system on a monitor.
- Interaction can also be more indirect, as in an intelligent building, which switches on the heating when you enter a room, or putting on a smartwatch.

What makes for a successful product?



IBM Simon Phone

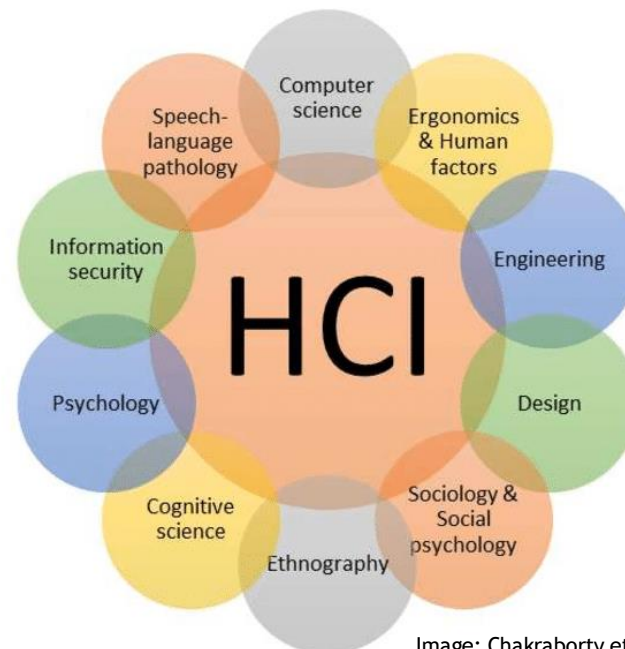


IBM, 1993

See Buxton collection
(MSR) for details

HCI and Human Factors

- Human Computer Interaction (HCI) is concerned with the design, evaluation and implementation of interactive computing systems for human use, and the study of phenomena surrounding them. [ACM]



Why study Human Computer Interaction?

- Help produce systems which are:
 - ***Useful***; accomplish what is required; play music, fly aircraft, format a document.
 - ***Usable***; do it easily and naturally, without danger of error, etc.
 - ***Used***; make people want to use it, be attractive, engaging etc.
- Many interesting jobs/roles.
- How do we do this?

Interdisciplinary nature of HCI HCI的跨学科性质

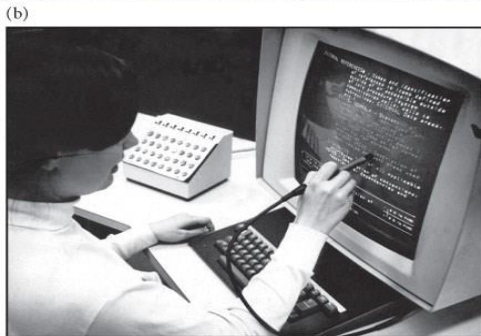
- HCI born out of interest in man-machine interaction following second world war (desire to produce more effective weapons systems), ergonomics.
- Addition of cognitive dimension, applied to any type of system. 认知维度的添加，适用于任何类型的系统
- Human Computer Interaction incorporates work on information science and technology.
- Draws on many areas, in computer science and systems design it is a central concern.

借鉴许多领域，在计算机科学和系统设计中，这是一个核心关注点。

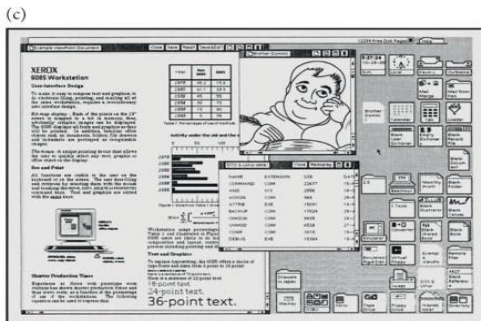
User Interface Technology



Sketchpad interactive system, Ivan Sutherland, 1964



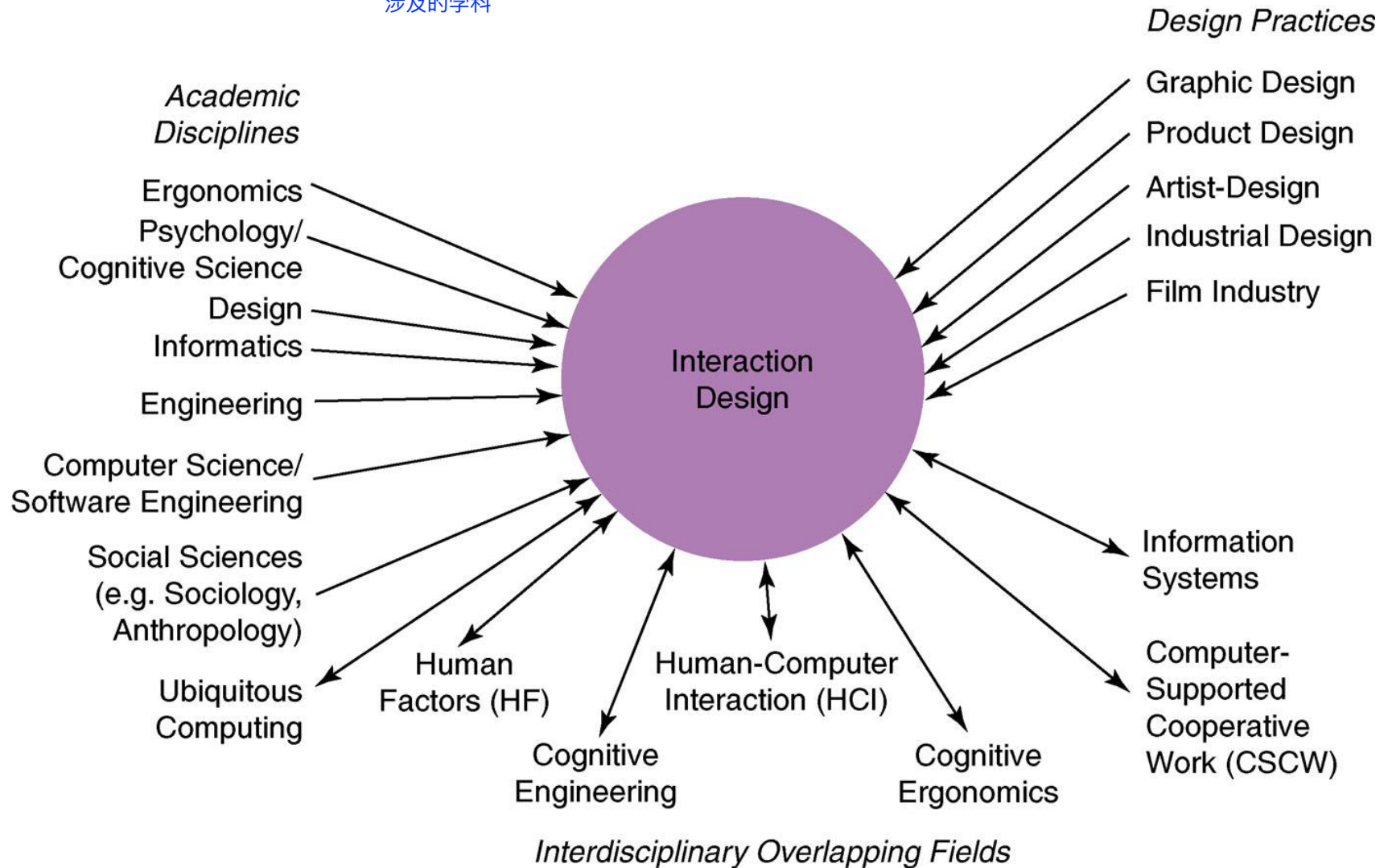
Hypertext Editing System, van Dam, Nelson, 1969



The Xerox Star graphical user interface 1981

Disciplines involved

涉及的学科



Usability

- If the user can't use it, it doesn't work. It's broken.
- Many systems are broken by design. They make “human error” highly likely, sometimes inevitable.
- If you're an engineer and you're producing systems which are poorly matched with the end users, their tasks, and their environment, you're not doing your job properly.



Interaction Design

- Need to take into account:
 - Who the users are
 - What activities are being carried out
 - Where the interaction is taking place
- Need to optimize the interactions users have with a product:
 - So that they match the users' activities and needs
- “Designing interactive products to support the way people communicate and interact in their everyday and working lives.”
 - Preece, Sharp and Rogers (2023)

Usability goals

- Effective to use
- Efficient to use
- Safe to use
- Have good utility
- Easy to learn
- Easy to remember how to use

User Experience

- How a product behaves and is used by people in the real world
 - The way people feel about it and their pleasure and satisfaction when using it, looking at it, holding it, and opening or closing it
 - “Every product that is used by someone has a user experience: newspapers, ketchup bottles, reclining armchairs, cardigan sweaters.” (Garrett, 2010)
 - “All aspects of the end-user's interaction with the company, its services, and its products. (Nielsen and Norman, 2014)
- Hassenzahl's (2010) model of the user experience
 - Pragmatic: how simple, practical, and obvious it is for the user to achieve their goals
 - Hedonic: how evocative and stimulating the interaction is to users

Hassenzahl (2010) 的用户体验模型

- 务实：用户实现目标是多么简单、实用和明显
- 享乐主义：互动对用户来说是多么令人回味和刺激

User experience goals

Desirable aspects

理想的方面

satisfying	令人满意的	helpful		fun	
enjoyable	令人愉悦的	motivating	有激励作用的	provocative	引发思考的 / 挑衅性的
engaging	引人入胜的 / 有吸引力的	challenging	有挑战性的	surprising	
pleasurable	使人愉快的	enhancing sociability	增强社交性的	rewarding	有成就感的 / 令人有收获的
exciting	令人兴奋的	supporting creativity	支持创造力的	emotionally fulfilling	
entertaining	娱乐性的 / 有趣的	cognitively stimulating			情感上充实的 / 情感满足的
			认知上有刺激性的 / 促进思考的		

Undesirable aspects

不良/负面方面

boring		unpleasant	令人不快的
frustrating	令人沮丧的 / 令人受挫的	patronizing	高高在上的 / 居高临下的(带有说教/轻视意味)
making one feel guilty		making one feel stupid	
annoying	令人恼火的	cutesy	过分可爱的 / 装可爱的(带贬义)
childish	幼稚的	gimmicky	花哨噱头式的 / 华而不实的

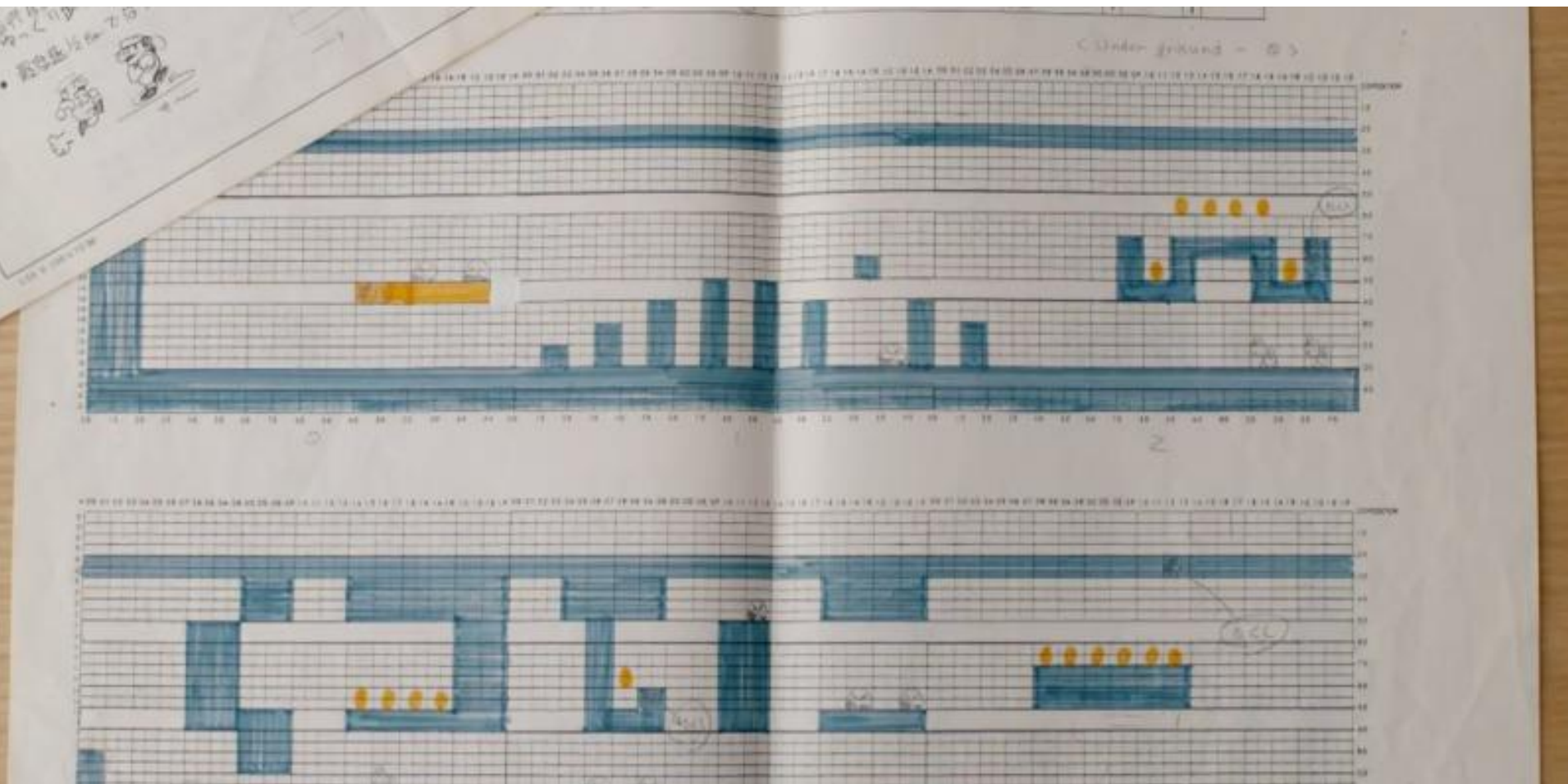
Bad user experiences

- It's simply not possible to do what you want to do.
- You can't work out how to do what you want to do.
- Working out how to do things takes a long time.
- Doing simple things takes longer than it should do.
- You regularly make the same error.
- You make an error which causes you to lose a lot of work/melts down the reactor.
- You find it frustrating to use.
- The system does something automatically which makes you irritated/angry.
- You think the system looks ugly.

Design example



Design example



Factors affecting design

- Human factors: “Know thy user”
 - Physical capabilities (motor skills, sight, hearing)
 - Cognitive capabilities (memory, attention span, decision making).
 - Social, cultural influences.
- Environmental factors
 - Lighting, ambient noise, operational conditions (eg. time pressure).

人为因素：“了解用户”

- 身体能力（运动技能、视觉、听觉）

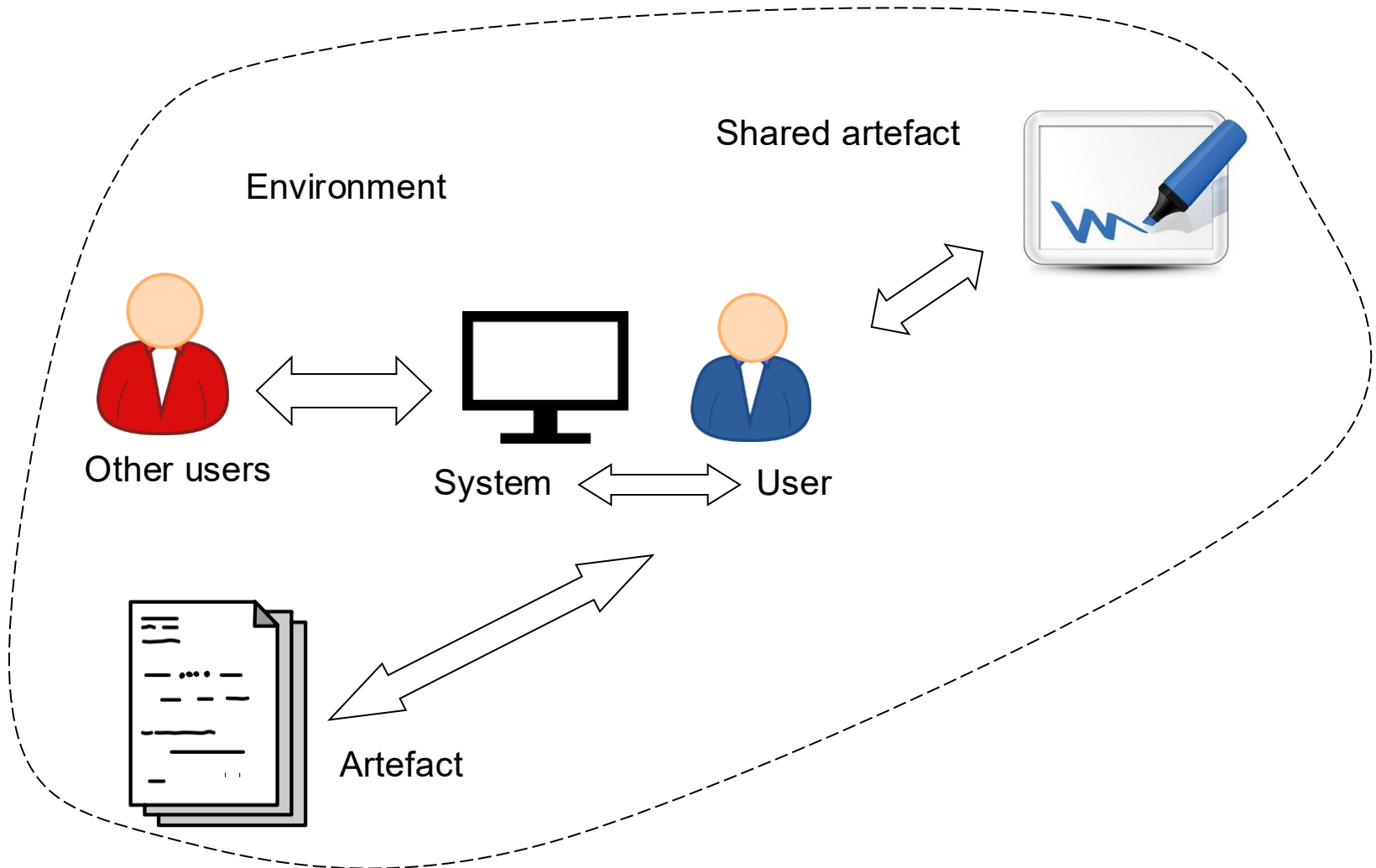
- 认知能力（记忆力、注意力、决策）。

- 社会、文化影响。

• 环境因素

— 照明、环境噪音、操作条件（例如时间压力）。

Interaction in Context



Users

- In the early days of computing, building interfaces for the devices was relatively easy:
 - Users very similar
 - Well-educated programmers
 - Similar tasks (ballistics, particle simulations).

类似的任务（弹道学、粒子模拟）。



Users

- In the modern era:
 - Different jobs (operator, manager)
 - Different tasks (design, administration, entertainment) 不同的任务 (设计、管理、娱乐)
 - Varying education (secondary, third level)
 - Different usages (casual, frequent).
 - Different age range
 - Different contexts



Understanding the user's needs

- Consider what might help people in the way they currently do things.
- Listen to what people want and get them involved.
- Understand what they do, and the environment they do it in.
- Use tried and tested user-based methods.

Dialogue

- In the design of many systems, interaction between the user and system takes the form of a dialogue (this is referred to as the **dialogue metaphor**).
- Messages are passed between user and system, these messages can be:
 - control
 - data
- On the system side:
 - prompt
 - data
 - status
 - error
 - help

Tasks and Goals

- Users are **goal directed**:
 - have specific problems to solve;
 - have desired outcomes to achieve;
 - feedback from interface must address this.
- Tasks are complex:
 - few relate solely to the computer;
 - decomposed into sub-tasks.
- For example:
 - task is to prepare a presentation;
 - sub-task to add a title slide.

What do HCI Practitioners do?

HCI从业者是做什么的

- **User Research:** Create an understanding of users and their activities, analysing tasks, collecting data about their activities, workplace studies, identify requirements, benchmark against competitors.
- **Design:** sketches, wireframes, mockups, prototypes.
设计：草图、线框、模型、原型。
• 评估：专家评估、实验室评估、现场评估。
- **Evaluate:** expert evaluations, lab evaluations, field evaluations.

Measures of Success

- What are the measurable factors:
 - Ease of learning, how much time and training is required for a new user to perform competently with the system.
 - Productivity measures, should support performance of tasks quickly and efficiently.
 - Error rates, error detection and error recovery, good feedback and awareness of situations by the user greatly aids.
 - User satisfaction - interviews and questionnaires.

成功衡量标准

•可衡量的因素是什么：

- 学习的便利性，新用户需要多少时间和培训才能熟练地执行系统。
- 生产力衡量标准，应支持快速有效地完成任务。
- 错误率、错误检测和错误恢复、良好的反馈和用户对情况的认识有很大帮助。
- 用户满意度-访谈和问卷。

Measures of Success

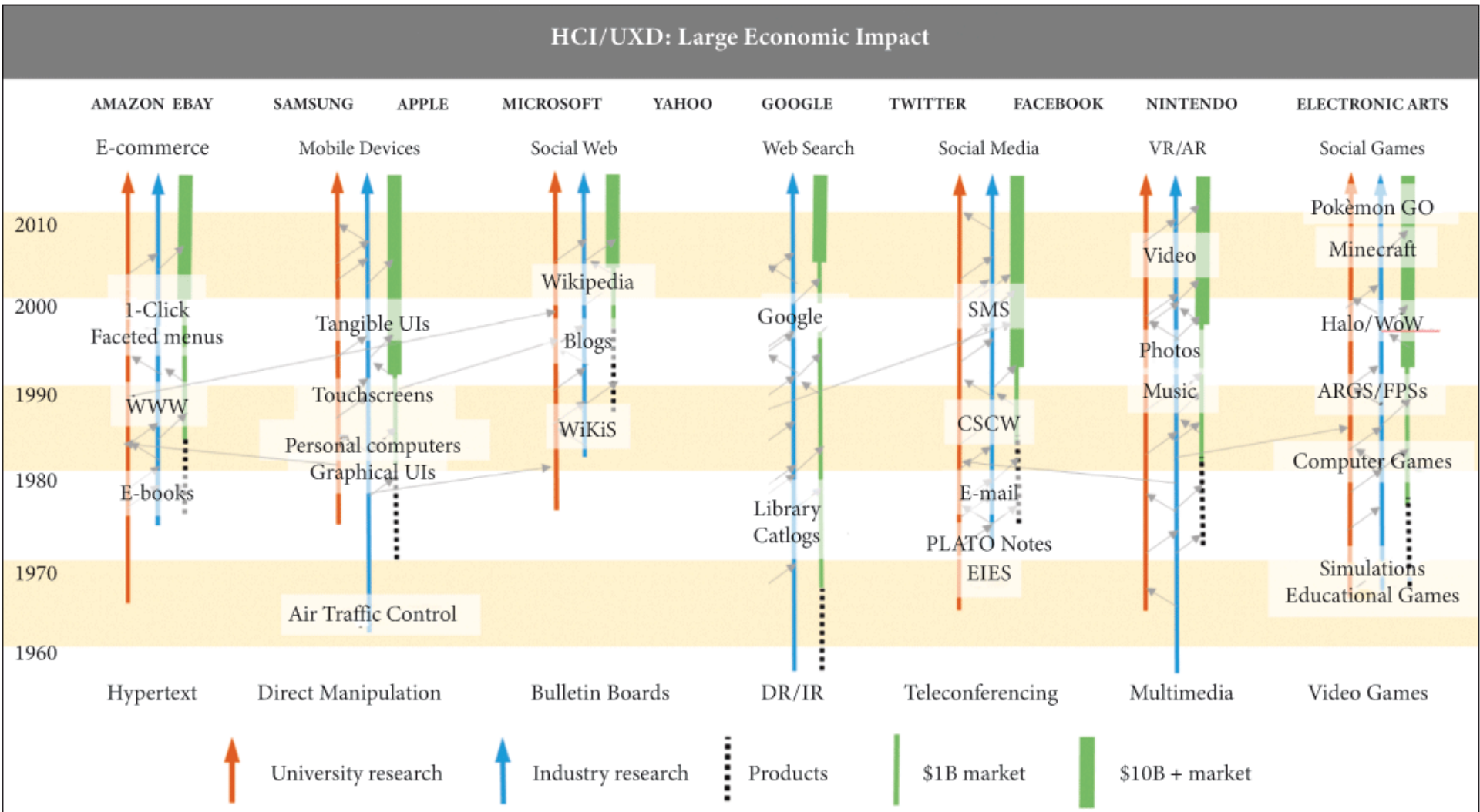
- Fiscal measures
 - Increased market share.
 - Reduced number of modifications.
 - Reduced need for customer support.

财政措施

—增加市场份额。

—减少修改次数。

—减少对客户支持的需求



Real world impact

- Then what?
 - Who is *actually* using the product?
 - How are they *actually* using it?
 - Task/artefact cycle.
- Ethics, unexpected (malicious?) uses

Real world impact

Apple and Google submit plan to fight AirTag stalking

Companies join forces to tackle unwanted tracking via Apple's gadget and similar devices such as Tile



📷 AirTags were released in 2021. Photograph: Apple Inc. Handout/EPA

Apple and **Google** are teaming up to thwart unwanted tracking through AirTags and similar gadgets.

The two companies behind the iPhone and the software that powers Android phones on Tuesday submitted a proposal to set standards for combatting secret surveillance on Bluetooth devices that were created to help people find lost keys, keep tabs on luggage or to locate other things that have a tendency to be misplaced.

'I log into a torture chamber each day': the strain of moderating social media

As jobs screening US, UK and EU platforms are shifted to India, viewing 'gore' to clean up the web has traumatised moderators



📷 Workers in the IT hub of Hyderabad in India said moderating sexual and abusive content had left them depressed, distressed and struggling to sleep. Photograph: Bloomberg/Getty Images

I had to watch every frame of a recent stabbing video ... It will never leave me," says Harun*, one of many moderators reviewing harmful online content in **India**, as social media companies increasingly move the challenging work offshore.

Develop your critical skills

培养你的关键技能

- Think of a system or device that you really like using. What's good about it?
- Think of a system that you really dislike using or which you find frustrating to use. What's wrong with it?
- The goal of the module is to help you build systems which deliver the first type of experience, and avoid the second.

Reflection

- Think of an app or system you have used recently.
 - What are the main design goals for the developers in terms of User Experience?
 - What aspects of human performance are relevant to this? Attention? Perception? Cognition? Memory? Error?
 - Are there any problems with the current design or nice design features addressing these issues?
- Write a half page document presenting your analysis, and bring to your group Tuesday. You will complete the other half at the lab.

Tutorial

- Within your group, each person should present their example, and group members discuss if they agree in terms of the User Experience goals of the developers.
- What aspects of human performance impact on the experience? Perception, attention, cognitive demands?(more on this at this week's lectures). (discuss)
- Are there any issues with the current design or nice features? What is good/bad about them in terms of human performance? (discuss)
- Add a short paragraph on the discussion and submit (individually) both your original paragraph and the discussion at the end of the lab.
- Keep track of time so that everyone gets to present theirs.